

Suryanshu Saini

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering — CGPA 8.78

Vellore, Tamil Nadu

Sep. 2022 – April 2026

WORK EXPERIENCE

UI/UX Design Intern

PhPPoets Pvt. Ltd.

May 2024 – July 2024

Udaipur, Rajasthan

- Created **7+ custom icon packs** and **3 responsive landing pages**, delivering pixel-perfect designs for enterprise clients across multiple industries
- Developed end-to-end mobile application designs including user research, wireframing, prototyping, and **usability testing** for 2 client projects
- Designed and iterated **11+ high-fidelity prototypes** using design thinking principles to enhance user experience and improve conversion rates
- Completed **two case studies** with direct feedback from client

PROJECTS

BCP | Rust, Typescript, Python, gRPC, HTTP

Dec 2025 – Present

- Proposed and implemented BCP (Better Context Protocol), a next-generation AI agent communication protocol using **Rust**, **gRPC**, and **Protocol Buffers**, achieving **10–100x serialization performance improvements** over existing solutions (MCP/JSON-RPC); patent pending.
- Built intent-based semantic tool discovery** using embedding search and context tiering (hot/warm/cold) to dynamically reduce LLM context usage, replacing static tool-dumping approaches in AI agent pipelines.
- Delivered a full cross-language **SDK ecosystem** (TypeScript, Python) with MCP backward-compatibility adapter, CLI tooling, VS Code extension, and **OpenTelemetry-based distributed tracing**; benchmarked end-to-end via automated test suite.

Cloud Clips | React Native, Expo, TypeScript, Convex, Bun

Sept 2025 – Present

- Engineered a **full-stack on-demand service marketplace** mobile application using **React Native** and **Expo**, supporting dual user roles (clients and barbers) with real-time booking, live chat, and push notifications across **iOS and Android**
- Architected **real-time backend** with **Convex** leveraging ACID transactions, automatic query caching, and WebSocket-free live sync, eliminating traditional backend infrastructure overhead while enabling instant availability updates across all connected clients
- Integrated **Stripe Connect** split-payment system with deferred payment capture, enabling automated platform commission collection and direct barber payouts with full refund and cancellation support

Hulaki | Go, Cobra, Fang, Bubbletea, TUI, API Testing, HTTP, GraphQL, gRPC

June 2025 – Present

- Developed high-performance terminal-native **API testing platform** using **Go** and modern CLI frameworks, delivering **3x faster** testing workflows compared to traditional GUI tools while reducing memory footprint by 80%
- Crafted intuitive **Terminal User Interface** with **Bubbletea** framework, featuring real-time syntax highlighting, interactive response visualization, and seamless keyboard navigation for enhanced developer experience
- Created modular CLI architecture using **Cobra** command framework and **Fang** HTTP client, enabling extensible plugin system

TECHNICAL SKILLS

Languages: Golang, Typescript, Python

Web Development: React, Next.js, SvelteKit, Hugo — Frontend Frameworks

Backend Technologies: Golang, Hono, Express, gRPC, REST APIs — API Development

Databases: PostgreSQL, Redis, Mongo — Database Management

DevOps & Tools: Docker, Git, GitHub Actions — CI/CD Pipeline

Automation Tools: n8n, Supabase, Playwright, Selenium

Cloud & Monitoring: AWS, Prometheus, Grafana — Cloud Computing

Protocols: GraphQL, WebSockets, WebRTC

UI/UX Design: Figma, Lottie — User Interface Design